# FARJANA FERDAUSI

**AI Engineer** | Machine Learning, Deep Learning, NLP, LLMs, RAG
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GitHub: https://github.com/farjanaferdausi-cs50ai | Portfolio: https://farjanaferdausi-cs50ai.github.io/Farjana-Ferdausi-Resume/
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## SUMMARY

AI Engineer building grounded, cloud-deployed LLM systems: RAG assistants, multi-agent workflows, data agents and computer-vision services using Python, Gemini, LangChain, LangGraph, FastAPI, Docker and Google Cloud. Fellow of the Google Cloud Gen AI Academy APAC Edition (Cohort 3) with 26 skill badges and 6 AI applications deployed on Cloud Run. Former HR professional with 14+ years of experience, bringing interview coordination and conduct, disciplined documentation, reporting and stakeholder communication to AI/ML engineering. Seeking AI Engineer roles owning systems from prototype to deployment.

## TECHNICAL SKILLS

**Languages & Data:** Python (pandas, NumPy, Matplotlib, Pydantic), SQL, REST APIs, Data pipelines / ETL
**Machine Learning:** scikit-learn, supervised and unsupervised learning, feature engineering, cross-validation, model evaluation
**Deep Learning, NLP & CV:** PyTorch, TensorFlow, Keras, Hugging Face Transformers, BERT fine-tuning, embeddings, OpenCV, YOLOv8, YOLOv11, object tracking (Deep SORT)
**LLMs & Generative AI:** RAG, hybrid retrieval (BM25 + embeddings), vector databases (ChromaDB), prompt engineering, structured outputs, tool calling, MCP, multi-agent systems, Gemini, LangChain, LangGraph, CrewAI, Google ADK, LangSmith, n8n
**Evaluation & Safety:** retrieval and answer evaluation sets, citation-grounded answers, confidence gating, refusal protocols, pytest, LangSmith, least-privilege IAM, human-in-the-loop approval
**MLOps & Cloud:** Docker, FastAPI, Git, CI/CD, Google Cloud (Vertex AI, Cloud Run, BigQuery, IAM, Secret Manager), Firebase (Auth, Firestore), Google Maps Platform, Google Earth Engine, Render, Vercel

## EXPERIENCE

**AI/ML Engineer — Independent | TR Enterprise (Start-up), Technology Division, Remote** 1 Sep 2026 – Present
- Design, build and deploy cloud-hosted RAG chatbots, computer-vision services and data-analysis agents using Gemini, Google ADK, LangChain, LangGraph, BigQuery, FastAPI and Docker, owning the ML lifecycle from prototype to deployment.
- Containerize FastAPI services with Docker and ship them to Google Cloud Run.
- Enforce safeguards in AI workflows: least-privilege IAM, read-only SQL with dry-run validation, and human approval before any write action.

**AI/ML Engineering Fellow | Google Cloud Gen AI Academy APAC Edition (Cohort 3)** 1 Aug 2026 – 2 Nov 2026
- Completed all three tracks plus Meet the Builders, the Ideathon and the AI Builder Cup 2026 Hackathon (completed; results 9 Nov 2026); earned a Certificate of Completion (Sep 2026) and an Ideathon Certificate of Participation (Nov 2026).
- Delivered 6 AI applications on Google Cloud Run: RAG Barista agent, BigQuery MCP data agent, AI Coffee Manager, Flood Risk Assistant Bangladesh, AI/ML Learning Journal and OmniClean AI.

**Artificial Intelligence Intern | CodeAlpha, India (Remote)** 20 Jul 2026 – 20 Aug 2026
- Delivered 4 end-to-end AI projects: a 20-language voice translator, a TF-IDF/cosine-similarity FAQ chatbot, an LSTM Bach chorale generator, and YOLOv8 + Deep SORT object tracking.
- Shipped each as a documented, runnable GitHub repository; received a Letter of Recommendation.

**HR Coordinator | Radisson Blu Dhaka Water Garden, Dhaka, Bangladesh** 15 Mar 2006 – 3 Oct 2020
- Screened resumes and candidates, scheduled and conducted interviews, and coordinated joining formalities.
- Administered leave and daily time and attendance through the HRIS, tracking late arrivals, early departures and absenteeism.
- Maintained employee files and HR records for 400+ employees with 100% documentation compliance; built Excel headcount and attrition reports.
- Coordinated employee engagement events; recognized with the Women's Loyalty Award (2018) and multiple Radisson Diamond, Platinum and Gold Awards.

**Career Break & Continuous Learning | Self-directed, Bangladesh** 4 Oct 2020 – Feb 2026
Planned, self-directed transition from HR to AI/ML engineering through structured, certified learning.
- Upskilled in HR: two Alison diplomas (Modern HRM, 2022; Human Resources, 2023) and three certificates (International & Strategic HRM; HR Policies & Organization Structure; HR in Food Service & Hospitality; 2022).
- Built GenAI foundations: AI Content Creation Bootcamp (Batch 82) and FB & YT Monetization (Batch 83), SR Dream IT, May – Jul 2025; Generative AI Mastermind (Outskill, 2025); Introduction to AI (Alison, 2026).
- Communication: Continuous English development, Keith Speaking Academy (2 Oct 2023 – Present), covering IELTS Speaking GOLD Interactive, Fluent Grammar, Vocabulary Vault and Idioms courses.

## PROJECTS

**OmniClean AI: Waste Intelligence Platform** | **AI Builder Cup 2026, Google Cloud & Hack2skill (Hackathon, Completed)**
1 Oct 2026 – 9 Nov 2026 (project submitted 18 Oct 2026)

- Built an AI waste-intelligence platform that identifies items in a photo with Gemini, gives sorting guidance in English and Bengali, blocks recycling advice for hazardous items (batteries, chemicals, sharps, pressurized cans) and flags scans below 0.6 confidence for retake.
- Added an impact dashboard (recycling rate, estimated CO2e avoided using EPA WARM v15 factors, 7-day trend) and a garment-factory monitor with a waste-per-1,000-pieces KPI and anomaly detection (> mean + 2 sigma) across 5 departments.
- Streamed analytics into 4 date-partitioned BigQuery tables using parameterized SQL and Application Default Credentials (no keys in source); kept the Gemini key server-side with a 4-model fallback ladder; deployed on Google Cloud Run.

**Tech: Gemini, Vertex AI, React, TypeScript, Express, Firebase, Firestore, BigQuery, Cloud Run**
**GitHub:** https://github.com/farjanaferdausi-cs50ai/OmniClean-AI

**ESBC-AI: Easy Service for Bangladeshi Citizens** | **Ostad Institute, Capstone Project (Module 27)** 19 Oct 2026 – 28 Oct 2026

- Built a bilingual (Bangla and English) RAG assistant giving citation-grounded answers on passport, NID, birth and death registration, fees, tax and social allowances, using only official .gov.bd pages stored with source URL and verification date.
- Curated a 29-document knowledge base across 7 service areas; designed hybrid retrieval (Gemini embeddings + BM25, reciprocal rank fusion) with ChromaDB and a LangGraph agent with a confidence gate (0.45 threshold) plus a NO_ANSWER refusal protocol.
- Evaluated on a fixed 29-question bilingual set (keyword-based): 100% top-5 retrieval hit rate, 100% answer accuracy and 100% correct refusal on out-of-scope questions; offline pytest suite covers retrieval, API contract and refusal path.
- Served via FastAPI with Gemini model fallback and response caching; containerized with Docker and deployed on Render (API) and Vercel (frontend).

**Tech: Python, FastAPI, LangGraph, RAG, Gemini embeddings, BM25, ChromaDB, Docker, Render, Vercel**
**GitHub:** https://github.com/farjanaferdausi-cs50ai/ESBC-AI | **Live:** https://esbc-ai.vercel.app

**AI/ML Learning & Career Journal** | **Google Cloud Gen AI Academy APAC Edition (Cohort 3), Ideathon** 22 Aug 2026 – 6 Sep 2026

- Developed an authenticated web app with multi-turn Gemini coaching, 7-part structured session summaries and a searchable timeline across four learning programs; added weak-skill detection, career-gap analysis and trend analysis over past sessions.
- Integrated Google Maps Platform (Places search, geolocation) for study-location tagging; secured data with Firebase Auth, owner-bound Firestore rules and Secret Manager (Gemini and Maps keys); deployed on Cloud Run.

**Tech: Gemini API, React, Node.js, Express, Firebase Auth, Firestore, Google Maps Platform, Secret Manager, Cloud Run**
**GitHub:** https://github.com/farjanaferdausi-cs50ai/AI-ML-Learning-Career-Journal-CloudRun

## EDUCATION & TRAINING

**AI & ML Engineer Bootcamp, Batch 6 | Ostad, Bangladesh 18 Feb 2026 – Present**

**AI Engineering & Data Science Bootcamp 4.0 | Codebasics, India 22 Jul 2026 – Present**

**Gen AI Academy APAC Edition (Cohort 3) | Google Cloud & Hack2skill 1 Aug 2026 – 2 Nov 2026**

**M.A., English Language & Literature | Bangladesh University (CGPA 3.52 / 4.00) 2013 – 2014**

**B.A., English Language & Literature | International Business Administration and Information System University (CGPA 3.25 / 4.00) 2008 – 2012**

## CERTIFICATIONS

- Google Cloud Gen AI Academy APAC Edition (Cohort 3), Certificate of Completion — Hack2skill & Google Cloud, Sep 2026 (ID 2026H2S09GCGENAIAPACC3-P00697). Verify: certificate.hack2skill.com
- Google Cloud Gen AI Academy APAC Edition (Cohort 3), Ideathon, Certificate of Participation — Hack2skill & Google Cloud, Nov 2026 (ID 2026H2S09GCGENAIAPACC3I-P00366). Verify: certificate.hack2skill.com
- Google Cloud Skills Boost — 26 skill badges (2026), including Gemini for App Developers, MLOps for Generative AI, Model Armor, Responsible AI, Introduction to AI Agents and Attention Mechanism
- CodeAlpha Artificial Intelligence Internship, with Letter of Recommendation — Aug 2026 (ID CA/DF1/205038)
- Generative AI Mastermind — Outskill, May 2025 & Jun 2026 (completed twice) | Advanced Prompt Engineering — Simplilearn, 2026
- Python for Data Science & ML — UniAthena, 2026 | Introduction to Artificial Intelligence — Alison, Feb 2026 (ID 25255027)
- Human Resources — Alison: Diploma in HR (Nov 2023), Diploma in Modern HRM (Jun 2022) and three short HR courses (2022)

## VOLUNTEERING & LEADERSHIP

- **SOS Children's Village | HR Team Representative, Radisson Blu | 2016 – 2020:** visited about once every two months to donate food, clothes, soap and funds, support the children's education and spend time with them.
- **Crisis Management Team Member | Radisson Blu | Aug 2015 – Oct 2020:** supported emergency preparedness; CPR and First Aid certified by Bangladesh Red Crescent Society (2009, refresher 2011); Fire and Safety trained with annual refreshers (2007–2019).
- **Social Committee Member | Radisson Blu | Dec 2015 – Oct 2020:** organized engagement activities and events.
- **Employee Child Care Center Supervisor | Radisson Blu | Oct 2009 – Oct 2020:** managed budget, hygiene inspections and daily activities for children aged 0–5.